glm-5-hf-model-card

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language:

- en
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GLM-5



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[Paper] [GitHub]

Introduction

We are launching GLM-5, targeting complex systems engineering and long-horizon agentic tasks. Scaling is still one of the most important ways to improve the intelligence efficiency of Artificial General Intelligence (AGI). Compared to GLM-4.5, GLM-5 scales from 355B parameters (32B active) to 744B parameters (40B active), and increases pre-training data from 23T to 28.5T tokens. GLM-5 also integrates DeepSeek Sparse Attention (DSA), largely reducing deployment cost while preserving long-context capacity.

Reinforcement learning aims to bridge the gap between competence and excellence in pre-trained models. However, deploying it at scale for LLMs is a challenge due to the RL training inefficiency. To this end, we developed slime, a novel **asynchronous RL infrastructure** that substantially improves training throughput and efficiency, enabling more fine-grained post-training iterations. With advances in both pre-training and post-training, GLM-5 delivers significant improvement compared to GLM-4.7 across a wide range of academic benchmarks and achieves best-in-class performance among all open-source models in the world on reasoning, coding, and agentic tasks, closing the gap with frontier models.

Benchmark

	GLM-5	GLM-4.7	DeepSeek-V3.2	Kimi K2.5	Claude Opus 4.5	Gemini 3 Pro	GPT-5.2 (xhigh)
HLE	30.5	24.8	25.1	31.5	28.4	37.2	35.4
HLE (w/ Tools)	50.4	42.8	40.8	51.8	43.4*	45.8*	45.5*
AIME 2026 I	92.7	92.9	92.7	92.5	93.3	90.6	-

	GLM-5	GLM-4.7	DeepSeek-V3.2	Kimi K2.5	Claude Opus 4.5	Gemini 3 Pro	GPT-5.2 (xhigh)
HMMT Nov. 2025	96.9	93.5	90.2	91.1	91.7	93.0	97.1
IMOAnswerBench	82.5	82.0	78.3	81.8	78.5	83.3	86.3
GPQA-Diamond	86.0	85.7	82.4	87.6	87.0	91.9	92.4
SWE-bench Verified	77.8	73.8	73.1	76.8	80.9	76.2	80.0
SWE-bench Multilingual	73.3	66.7	70.2	73.0	77.5	65.0	72.0
Terminal-Bench 2.0 (Terminus 2)	56.2 / 60.7 †	41.0	39.3	50.8	59.3	54.2	54.0
Terminal-Bench 2.0 (Claude Code)	56.2 / 61.1 †	32.8	46.4	-	57.9	-	-
CyberGym	43.2	23.5	17.3	41.3	50.6	39.9	-
BrowseComp	62.0	52.0	51.4	60.6	37.0	37.8	-
BrowseComp (w/ Context Manage)	75.9	67.5	67.6	74.9	67.8	59.2	65.8
BrowseComp-Zh	72.7	66.6	65.0	62.3	62.4	66.8	76.1
τ^2-Bench	89.7	87.4	85.3	80.2	91.6	90.7	85.5
MCP-Atlas (Public Set)	67.8	52.0	62.2	63.8	65.2	66.6	68.0
Tool-Decathlon	38.0	23.8	35.2	27.8	43.5	36.4	46.3
Vending Bench 2	$4,432.12	$2,376.82	$1,034.00	$1,198.46	$4,967.06	$5,478.16	$3,591.33

*: refers to their scores of full set.

†: A verified version of Terminal-Bench 2.0 that fixes some ambiguous instructions. See footnote for more evaluation details.

Footnote

- **Humanity's Last Exam (HLE) & other reasoning tasks**: We evaluate with a maximum generation length of 131,072 tokens (`temperature=1.0, top_p=0.95, max_new_tokens=131072`). By default, we report the text-only subset; results marked with * are from the full set. We use GPT-5.2 (medium) as the judge model. For HLE-with-tools, we use a maximum context length of 202,752 tokens.
- **SWE-bench & SWE-bench Multilingual**: We run the SWE-bench suite with OpenHands using a tailored instruction prompt. Settings: `temperature=0.7, top_p=0.95, max_new_tokens=16384`, with a 200K context window.
- **BrowserComp**: Without context management, we retain details from the most recent 5 turns. With context management, we use the same discard-all strategy as DeepSeek-v3.2 and Kimi K2.5.
- **Terminal-Bench 2.0 (Terminus 2)**: We evaluate with the Terminus framework using `timeout=2h, temperature=0.7, top_p=1.0, max_new_tokens=8192`, with a 128K context window. Resource limits are capped at 16 CPUs and 32 GB RAM.
- **Terminal-Bench 2.0 (Claude Code)**: We evaluate in Claude Code 2.1.14 (think mode, default effort) with `temperature=1.0, top_p=0.95, max_new_tokens=65536`. We remove wall-clock time limits due to generation speed, while preserving per-task CPU and memory constraints. Scores are averaged over 5 runs. We fix environment issues introduced by Claude Code and also report results on a verified Terminal-Bench 2.0 dataset that resolves ambiguous instructions (see: https://huggingface.co/datasets/zai-org/terminal-bench-2-verified).
- **CyberGym**: We evaluate in Claude Code 2.1.18 (think mode, no web tools) with (`temperature=1.0, top_p=1.0, max_new_tokens=32000`) and a 250-minute timeout per task. Results are single-run Pass@1 over 1,507 tasks.

- **MCP-Atlas**: All models are evaluated in think mode on the 500-task public subset with a 10-minute timeout per task. We use Gemini 3 Pro as the judge model.
- **τ²-bench**: We add a small prompt adjustment in Retail and Telecom to avoid failures caused by premature user termination. For Airline, we apply the domain fixes proposed in the Claude Opus 4.5 system card.
- **Vending Bench 2**: Runs are conducted independently by Andon Labs.

Serve GLM-5 Locally

Prepare environment

The following open-source frameworks support local deployment of GLM-5:

- vLLM (v0.19.0+)
- SGLang (v0.5.10+)
- KTransformers (v0.5.3+)
- Transformers (v0.5.4+)
- xLLM (v0.8.0+)

Deploy

- vLLM

  ```
  vllm serve zai-org/GLM-5 \
       --tensor-parallel-size 8 \
       --gpu-memory-utilization 0.85 \
       --speculative-config.method mtp \
       --speculative-config.num_speculative_tokens 3 \
       --tool-call-parser glm47 \
       --reasoning-parser glm45 \
       --enable-auto-tool-choice \
       --served-model-name glm-5
  ```

  Check the recipes for more details.

- SGLang

  ```
  sglang serve \
    --model-path zai-org/GLM-5 \
    --tp-size 8 \
    --tool-call-parser glm47  \
    --reasoning-parser glm45 \
    --speculative-algorithm EAGLE \
    --speculative-num-steps 3 \
    --speculative-eagle-topk 1 \
    --speculative-num-draft-tokens 4 \
    --mem-fraction-static 0.85 \
    --served-model-name glm-5
  ```

  Check the sglang cookbook for more details.

- xLLM and other Ascend NPU

  Please check the deployment guide here.

- KTransformers

  Please check the deployment guide here.

Citation

If you find GLM-5 useful in your research, please cite our technical report:

```bibtex
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